Conditional DiffTraj on Porto taxis and Turin buses

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Question and data

Given the attributes of a long taxi journey, how closely do trajectories sampled by DiffTraj reproduce the lengths and spatial distribution of observed journeys? Zhu et al. studied taxi GPS trajectories from Chengdu and Xi'an in their main experiment and reported an additional Porto comparison in Appendix C, Table 5, where DiffTraj was compared with VAE, TrajGAN, DiffWave and Diff-LSTM on four distributional measures. The Porto experiment reproduces the published benchmark under a documented long-trip rule. A second experiment applies the same generator to observed Turin bus journeys.

The UCI Porto taxi trajectory corpus (source linked below) records taxi positions at 15-second intervals. Each row describes a journey with a taxi identifier, departure timestamp and ordered longitude–latitude polyline. From 1,710,670 rows, we retained complete, finite tracks containing at least 120 original positions before interpolation. At the native cadence, 120 positions span 29 min 45 s. The resulting 55,267 journeys from 439 taxis represent the long-trip population. The paper describes a 120-point rule in Appendix B.1, but gives no separate retained count or exact split for Porto.

We partitioned by taxi identity to keep vehicles disjoint. The training partition contained 46,470 journeys, the development partition 5,241 and the reserved partition 3,556. We used 512 fixed development requests for checkpoint selection and sampled 3,000 reserved journeys for the final comparison.

Model and training

DiffTraj represents each journey as two coordinate channels interpolated to 200 positions. During training, Gaussian noise is added to an observed trajectory. A temporal U-Net takes the noisy coordinates, the noise level and a trip header and predicts the injected noise by minimizing mean squared error. The header specifies departure-time bin, distance, duration, original point count, average displacement, average speed, and origin and destination cells. Sampling begins with Gaussian noise and iteratively denoises it. We restore the original sample count before scoring. Supplying distance and endpoints makes this an evaluation of conditional generation.

We trained the released coordinate architecture from scratch for 200 epochs with 500 noise levels, 100 reverse sampling steps and global batch size 1,024 on four NVIDIA L20 GPUs (256 journeys per GPU). Inputs were normalized from training data. The authors' published training loop uses fixed-rate AdamW and saves ordinary network weights every ten epochs. Following instability in an earlier local fit, the two runs reported here used AdamW, gradient clipping at norm 1 and cosine learning-rate decay from 0.0002 to 0.00002. Epochs 193 and 179 minimized fixed development noise loss for seeds 42 and 43, respectively. The second fit changed initialization, training order and training noise while retaining the data, validation rule, sampling noise and evaluation requests.

Results against the published Porto benchmark

Density and OD errors compare point occupancy and ordered origin–destination cell frequencies on a 16×16 grid. Length error compares distributions of cumulative journey distance. All three use Jensen–Shannon divergence (JS), with zero for identical distributions. Our length histogram uses 20 training-quantile bins. Our pattern F1 measures the overlap of the 20 most frequently visited cells.

	Density JS ↓	OD JS ↓	Length JS ↓	Pattern F1 ↑
Published Table 5	.00870	.01320	.02420	.847
Long trips, seed 42	.01357	.00594	.00635	.750
Long trips, seed 43	.00944	.00698	.00395	.800

How the grid changes hotspot F1. We scored each set of 3,000 draws against identical observed journeys under three grid definitions. Each grid divides the specified longitude and latitude ranges into 16 equal intervals per axis. We count GPS points in each cell and rank the cells separately for observed and generated trips. Both hotspot sets contain 20 cells, so F1 equals the number of shared cells divided by 20.

Training coordinates used to set grid edges	Seed 42: hits / F1	Seed 43: hits / F1
Long trips, full min–max (primary)	15/20: 0.750	16/20: 0.800
All eligible lengths sampled for training, full min–max	17/20: **0.850**	16/20: 0.800
Same sample, 1st–99th percentile per axis	15/20: 0.750	16/20: 0.800

For seed 42 the shared full-range grid places two additional cells in the overlap. Seed 43 has 16 overlapping cells on every grid. Only the scoring boundaries change; all three rows evaluate the long-trip model. Full-range grids clip outlying points to the edge; the percentile grid excludes overflow from hotspot ranking. The shared-grid score is close to the published 0.847, whose Porto boundaries and hotspot count remain unspecified.

What does the length score mean in kilometres? The observed median was 13.96 km, compared with 12.39 and 12.64 km for seeds 42 and 43. Across requests, generated length averaged 2.01 and 1.51 km below observed length. The first Wasserstein distance between the distributions was 2.28 and 1.92 km, respectively: this measures the average separation of corresponding length quantiles. A small binned JS therefore coexists with a measurable shortfall in distance.

Conclusion. We reproduced the conditional training and sampling procedure on public Porto GPS data under an explicit long-trip rule. Both fits generated shorter median trips and recovered 15 to 17 of the 20 observed hotspots across the specified grids. The scope of this result is journeys of at least 120 recorded positions.

Turin: collecting vehicle positions and reconstructing trips

The Turin transport operator, Gruppo Torinese Trasporti (GTT), publishes a live snapshot of vehicle positions. To build a journey history, a Python collector on the research server requested the GTT vehicle-position feed, processed the response and waited 60 seconds before the next request. Each response was a binary GTFS Realtime Protocol Buffers message containing reports from multiple vehicles. The collector archived each distinct nonempty payload as a compressed `.pb.gz` file and decoded the position reports into SQLite.

Observation format. One SQLite row represents one vehicle fix. The fields below are listed in storage order.

Field	Meaning
`vehicle, trip, service_date, route`	Vehicle ID, scheduled trip ID, service date and route ID.
`t`	Vehicle-reported Unix timestamp in seconds. Fetch time was recorded separately for each poll.
`lat, lon`	Latitude and longitude in decimal degrees.
`speed`	Optional reported speed in m/s. Empty for all fixes on this service day.

Repeated snapshots could contain the same fix. The key (vehicle, trip, service date, vehicle timestamp) prevented double counting. From 02:47 on 12 November to 02:46 on 13 November 2026 (Turin time), 1,393 polls yielded 354,114 distinct fixes from 643 vehicles on 94 routes. Of those polls, 110 returned no vehicles. The median interval between successive observations was 63 seconds.

Journey reconstruction. Fixes were grouped by vehicle, service date and trip, then sorted by vehicle timestamp. GTT's static GTFS timetable supplied the active service, route type and scheduled terminals. A new segment began at a gap over 180 seconds, a nonpositive time interval or implied speed over 35 m/s. Retained segments contained at least ten fixes, ten minutes and 500 m of travel. Each endpoint had to lie within 200 m of its scheduled terminal and within 15 minutes of its scheduled time; observed duration had to cover at least 70% of the scheduled duration. Checking bus routes against the operator's fleet catalogue yielded 3,262 bus journey candidates.

Analysis and model formats. The cleaned gzip-compressed CSV retained one row per fix, with segment ID, vehicle mode, fleet and terminal-check fields. Its interval speed is the great-circle distance between successive fixes divided by their actual timestamp difference. For training, each segment became an ordered longitude–latitude sequence in a compressed NumPy archive (`.npz`), with vehicle ID, start time, elapsed duration and original point count. A median bus journey contained 46 reported positions. Interpolation to 200 model positions was applied only after reconstruction.

Turin buses: training and evaluation

Splitting by vehicle gave 2,555 training, 366 development and 341 reserved journeys. Both fits used the Porto network and eight trip attributes, measured elapsed duration and local departure time. Only buses entered the experiment; no vehicle-class input was supplied. At global batch 1,024, 200 epochs on Turin data give 600 optimizer updates. Reducing the batch to 56 gives 9,200 updates, matching the Porto count. The smaller batch also changes gradient variability. The two training seeds used identical evaluation requests and sampling noise.

Turin, 341 reserved requests	Density JS	OD JS	Length JS	Hotspot F1
Batch 1,024, seed 42	.0607	.6513	.1387	.85
Batch 1,024, seed 43	.4931	.6889	.3718	.15
Batch 56, seed 42	.0120	.3187	.0515	.95
Batch 56, seed 43	.0171	.3375	.0611	.95
Observed development vs. reserved	.0587	.3136	.0895	.75

Both smaller-batch models recovered 19 of the 20 observed hotspots. Generated median lengths were 10.01 and 10.34 km against 11.13 km observed. At seed 43 the fraction of generated points outside the training bounds fell from 70.41% to 0.36%. Length W1 fell from 8.86 to 0.54 km at seed 42 and from 95.80 to 0.51 km at seed 43.

For an equal-size comparison, 50 subsets of 341 Porto requests gave median length JS of .0157 and .0121 and OD JS of .0124 and .0126 for seeds 42 and 43. Turin showed larger errors on both measures. The observed Turin OD divergence of .3136 between disjoint vehicle groups illustrates the variation in route coverage within this service day. Later collection dates are reserved for a separate assessment of generalization.

Sources

Zhu et al., DiffTraj, NeurIPS 2023, especially Appendices B.1, B.2 and C.1.
UCI Porto taxi trajectory corpus.
Author implementation and local evaluation records.

Turin observations: GTT vehicle positions and static GTFS. Turin paired evaluation records.

What the Turin scores measure

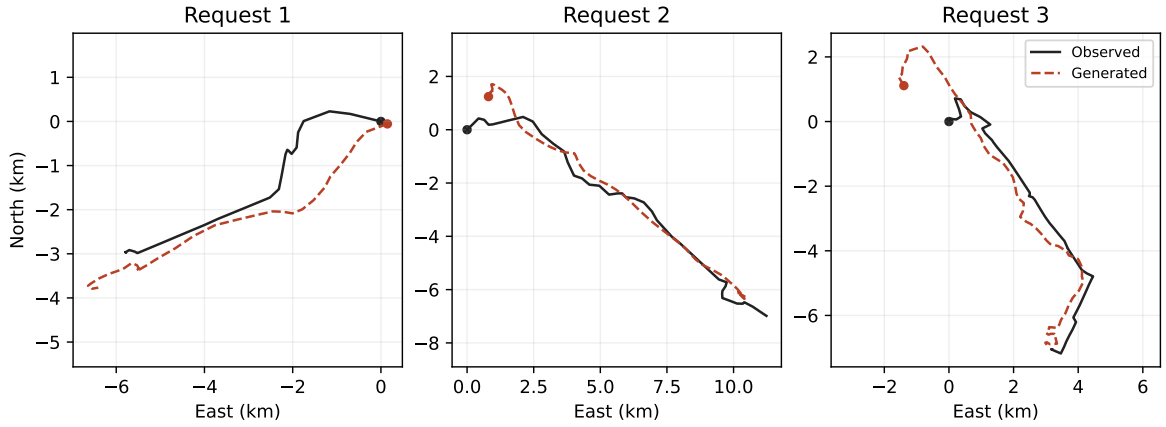


Figure 1. The first three reserved requests in the fixed evaluation order. Observed and generated positions (batch 56, seed 42) share local axes relative to the observed start. Dots mark departures. Connecting consecutive GPS fixes shows the sampled path at the reporting cadence.

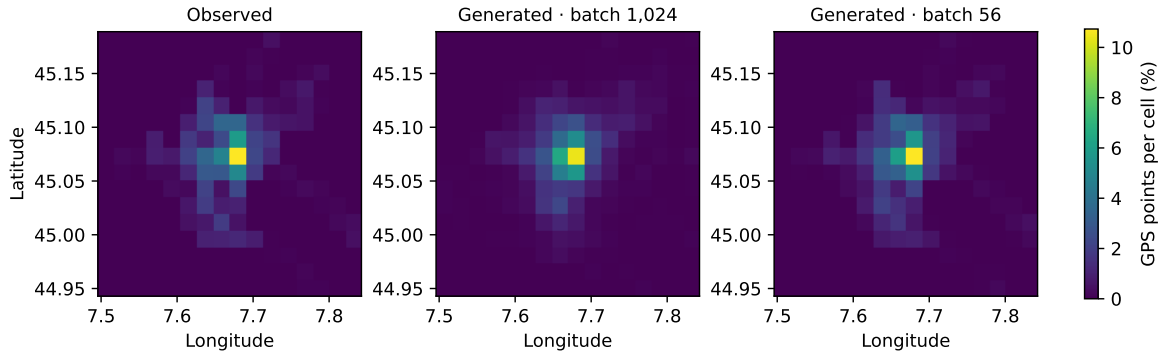


Figure 2. Point occupancy across all 341 requests at seed 42. Each panel uses the same 16 by 16 training grid and colour scale. The full-batch comparison in the table shows why hotspot overlap alone can conceal errors in origin–destination pairs.

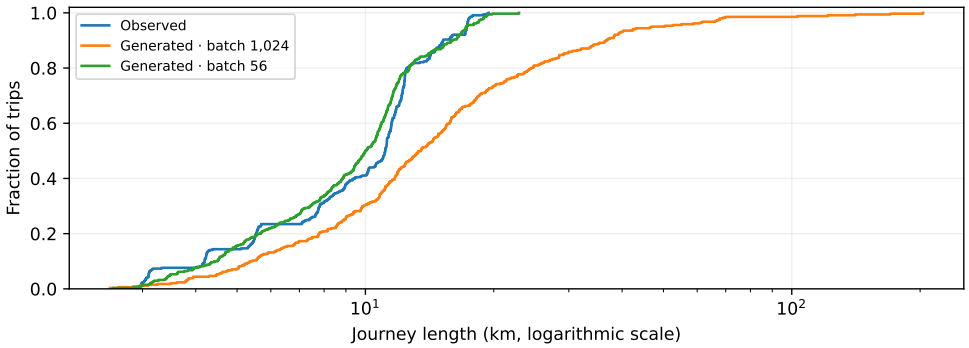


Figure 3. Empirical journey-length distributions for the same requests and seed. The logarithmic distance axis includes the upper tail. At batch 56 the generated median is closer to the observed median.